

Problems for week 9

(Questions on this problem sheet are NOT assessed)

Problem 1. Let C denote the triangle with vertices $(0,0)$, $(1,0)$ and $(1,2)$, oriented counterclockwise. Evaluate the line integral

$$\oint_C xy \, dx + x^2 y^3 \, dy$$

by two methods:

- (a) directly;
- (b) using Green's Theorem.

Problem 2. Let $\mathbf{F} = (y - \cos y, x \sin y)$ and let C be the circle $(x - 3)^2 + (y + 4)^2 = 4$ oriented clockwise. Use Green's Theorem to evaluate the line integral

$$\oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Problem 3. If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, then $f\mathbf{F}$ is the vector field defined by

$$(f\mathbf{F})(x, y) := f(x, y)\mathbf{F}(x, y).$$

Assume that the appropriate partial derivatives exist and are continuous. Show that

$$\operatorname{div}(f\mathbf{F}) = f \operatorname{div}(\mathbf{F}) + \mathbf{F} \cdot \nabla f.$$

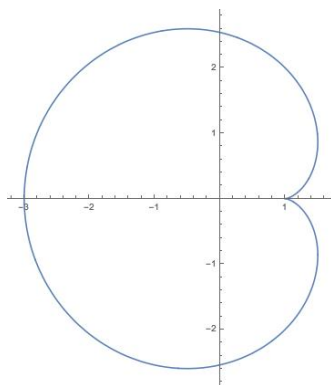


FIGURE 1. The cardioid parametrised by $x(t) = -\cos(2t) + 2 \cos t$ and $y(t) = -\sin(2t) + 2 \sin t$, for $0 \leq t \leq 2\pi$.

Problem 4. Compute the area of the region enclosed by the *cardioid* (see Fig. 1) parametrised by

$$C : (x(t), y(t)) = (-\cos(2t) + 2\cos t, -\sin(2t) + 2\sin t), \quad 0 \leq t \leq 2\pi.$$

A function $g : R \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is called *harmonic* on R if it satisfies Laplace's equation $\Delta g = 0$ on R , where

$$\Delta g := \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2}.$$

Problem 5. Use Green's Theorem in Normal Form to prove **Green's first identity**:

$$\iint_R f \Delta g \, dA = \oint_C f(\nabla g) \cdot \hat{n} \, ds - \iint_R \nabla f \cdot \nabla g \, dA,$$

where R and C satisfy the hypotheses of Green's Theorem and the appropriate partial derivatives of f and g exist and are continuous. (The quantity $\nabla g \cdot \hat{n} = D_{\hat{n}}g$ occurs in the line integral. This is the directional derivative in the direction of the normal vector $\hat{n}$ and is called the *normal derivative* of g .)

Problem 6. Use Green's first identity to prove **Green's second identity**:

$$\iint_R (f \Delta g - g \Delta f) \, dA = \oint_C (f \nabla g - g \nabla f) \cdot \hat{n} \, ds$$

where R and C satisfy the hypotheses of Green's Theorem and the appropriate partial derivatives of f and g exist and are continuous.